

分析流体的运动有两种方法: differential

- ① 试图描述流场中每一点详细流动状态
- ② 取一有限区域, 通过进出该区域的流量平衡, 来求该总的流动效果, 例如作用于流体的力或转矩, 或者是所有的能量交换。

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Control Volume (控制体积法)

Investigating steps,

1. develop the Concept of control volume.
2. find the rate of change of an arbitrary gross fluid property, Reynold transport theorem.
3. deriving 4 basic control-volume relations of fluid mechanics

mass
linear momentum
angular momentum
on
4. a special case of frictionless, shaft-work-free momentum and energy: Bernolli equation

§ 3.1 Basic Physical Laws of F. M.

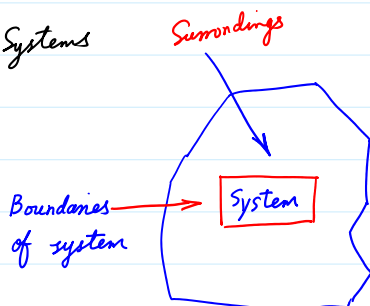
流体力学问题在何者看来而对变化的流动状态进行几何分析, 边界分析以及力学分析。

对任意流动问题的分析方法包括:

1. 控制体积, 或大尺度分析 (Chapter 3)
2. 微分法, 小尺度分析 (Chapter 4)
3. 实验或量纲分析 (chapter 5)
(dimensional analysis)

从工程上来讲, 控制体积法更为重要。尽管比较粗糙, 但总是很有效。

1. Systems



① the system is a fixed quantity of mass, m

" m " is conserved and does not change.

Conservation of mass:

$$m_{\text{sys}} = \text{Const}$$

$$dm/dt = 0$$

② If the surroundings exert a net force F on the system, Newton's second law

states, $\vec{F} = m\vec{a} = m \cdot \frac{d\vec{V}}{dt} = \frac{d(m\vec{V})}{dt}$

Recall (2.12)

$$\rho a = \sum f = -\nabla p + \rho g + \mu \nabla^2 \vec{V}$$

(Viscous incompressible fluid)

粘性不可压缩流体

* 在流体力学中, 牛顿运动定律被称为“动量关系式”

linear momentum

需要注意的是: 这个矢量定律可由三个标量方程表示

$$\begin{cases} F_x = ma_x \\ F_y = ma_y \\ F_z = ma_z \end{cases} \quad \text{scalar}$$

③ If the surroundings exert a net moment M about the center of mass of the system, there will be a rotation effect,

$$\vec{M} = \frac{d\vec{H}}{dt}$$

$$\vec{H} = \sum (\vec{r} \times \vec{V}) \delta m \quad (\text{矢量积})$$

$$\left(\begin{array}{l} \text{叉乘: } \vec{c} = \vec{a} \times \vec{b} \\ \text{"右手法则"} \end{array} \right) \quad \left\{ \begin{array}{l} |\vec{c}| = |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin(\angle \vec{a}, \vec{b}) \end{array} \right.$$

Angular momentum relation

$\vec{H}$ 是系统关于其重心的角动量.

$$\begin{cases} M_x = dH_x/dt \\ M_y = dH_y/dt \\ M_z = dH_z/dt \end{cases}$$

对于任意质量 m 和动量 M , H 是非标量守恒的.

它包括了9项, 在这里我们一般处理为“

绕一固定 x 轴旋转的刚体, 因此上述方程可简化为:

$$M_x = I_x \frac{d}{dt} (\omega_x) \quad (3.4)$$

ω_x —— 物体的角速度

I_x —— 关于 x 轴的惯性动量

但实际上, 流体系统并不是刚性的, 所以很少能简化到 (3.4) 这种关系式。

④ If heat dQ is added to the system or work dW is done by the system, the system energy dE must change according to First Law of thermodynamics

$$dQ - dW = dE$$

$$\frac{dQ}{dt} - \frac{dW}{dt} = \frac{dE}{dt}$$

While the second law of thermodynamics relates entropy change ds to heat added dQ and absolute temperature T :

$$ds \geq \frac{dQ}{T}$$

除上述方程式和热力学性质外, 还需要补充

状态方程 $p = p(p, T)$ 和 内能 $e = e(p, T)$

* 本章的目的就是将4种守恒律应用于

适合流体中任意区域的控制体。

1. 质量守恒方程
2. 线性动量方程
3. 角动量方程
4. 能量方程

此外, 尚需要补充一状态方程, 例如理想气体

状态方程 (Ideal gas / perfect gas law)

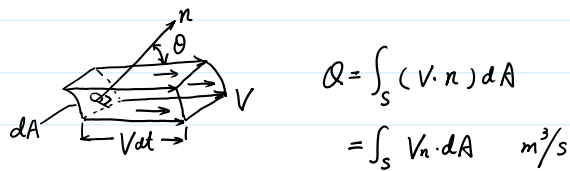
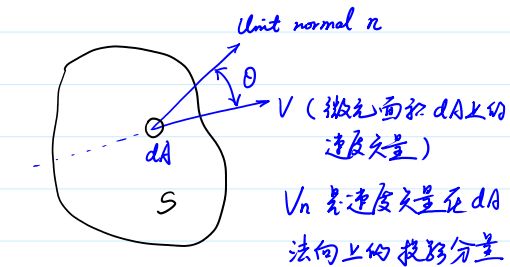
2. Volume and Mass Rate of Flow

单位时间内通过流场中特定面积的流体量称为

流量。通常用体积或质量来度量所通过的

流体量, 相应地就有体积流量 Volume flow rate.

和质量流量 Mass flow rate 的概念.



这里引入点积 $\mathbf{V} \cdot \mathbf{n}$ 是为了区别流量 Q 是向内流, 还是向外流.

通常我们认为 $\mathbf{n}$ 是向外的法向单位矢量

如果 $\mathbf{V} \cdot \mathbf{n}$ 为正值 $\Rightarrow$ 表明是向外流

$\mathbf{V} \cdot \mathbf{n}$ 为负值 $\Rightarrow$ 表明是向内流

Mass flow in equals to Volume flow

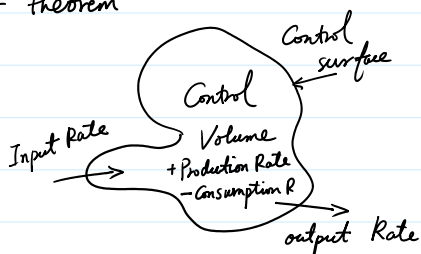
times by density.

$$\dot{m} = \int_S \rho (\mathbf{V} \cdot \mathbf{n}) dA = \int_S \rho V_n dA$$

$$\dot{m} = \rho Q \quad (\text{if } \rho = \text{const})$$

§ 3.2 The Reynolds Transport Theorem

Reynolds transport theorem can be stated as



what was already there + what goes in - what comes out

= what is there

雷诺定理是用来表示质量的守恒定律

§ 3.3 Continuous Relation and Mass conservation equation

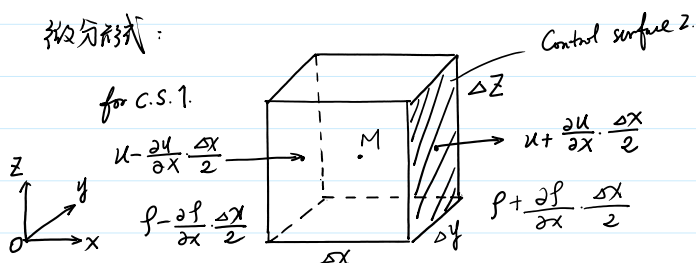
1. System here in fluid mechanics

这里所研究的系统: 是由特定流体质点组成的微团.

随之质点的运动，系统的位置和形状都会发生改变，但是它始终是由同一组流体质点所组成。

2. 流体的运动存在一些内在规律，其中之一就是质量守恒定律，而连续性方程就是质量守恒定律在流体力学中具体的数学描述，也叫做质量守恒方程。

微分形式：



取一平行六面体控制体， $M \rightarrow$ 形心

M 的速度表达为 (u, v, w) ，密度为 ρ

在左右控制面的形心点上：the velocity component in x direction,

$$\text{left side: } u - \frac{\partial u}{\partial x} \frac{\Delta x}{2},$$

$$\text{right side: } u + \frac{\partial u}{\partial x} \frac{\Delta x}{2}$$

the density of fluid is, left side: $\rho - \frac{\partial \rho}{\partial x} \frac{\Delta x}{2}$

$$\text{right side: } \rho + \frac{\partial \rho}{\partial x} \frac{\Delta x}{2}$$

Since the area of control surface is considerably small,

thus, the above value are taken as their average quantities of whole area.

由于控制面面积很小，以上速度和密度可付为它们各自控制面上的平均值。

则净流入控制体的质量流量为

$$\begin{aligned} & \underbrace{\left(\rho - \frac{\partial \rho}{\partial x} \frac{\Delta x}{2} \right) \left(u - \frac{\partial u}{\partial x} \frac{\Delta x}{2} \right) \Delta y \Delta z}_{\text{inflow}} - \underbrace{\left(\rho + \frac{\partial \rho}{\partial x} \frac{\Delta x}{2} \right) \left(u + \frac{\partial u}{\partial x} \frac{\Delta x}{2} \right) \Delta y \Delta z}_{\text{outflow}} \\ &= \left(\rho u - \rho \frac{\partial u}{\partial x} \frac{\Delta x}{2} - u \frac{\partial \rho}{\partial x} \frac{\Delta x}{2} + \frac{\partial \rho}{\partial x} \frac{\partial u}{\partial x} \frac{\Delta x^2}{4} \right) \Delta y \Delta z - \\ & \quad \left(\rho u + \rho \frac{\partial u}{\partial x} \frac{\Delta x}{2} + u \frac{\partial \rho}{\partial x} \frac{\Delta x}{2} + \frac{\partial \rho}{\partial x} \frac{\partial u}{\partial x} \frac{\Delta x^2}{4} \right) \Delta y \Delta z \\ &= - \frac{\partial(\rho u)}{\partial x} \Delta x \Delta y \Delta z \end{aligned}$$

以此类推：沿 y, z 方向净流入控制体的质量流量分别为：

$$- \frac{\partial(\rho v)}{\partial y} \Delta x \Delta y \Delta z ; - \frac{\partial(\rho w)}{\partial z} \Delta x \Delta y \Delta z$$

$$\frac{\partial \rho}{\partial t} \Delta x \Delta y \Delta z ; - \frac{\partial \rho}{\partial z} \Delta x \Delta y \Delta z$$

Because the control volume is fixed (控制体的体积不变)
而且流体质点是连续分布的, 因此如果从全部控制面净流入
的质量流量不为零, 那么控制体内流体的密度一定发生变化.

这里假设选取的控制体足够小, 其中心点的密度 ρ , 可以
作为控制体内流体的平均密度 (忽略几何效应)

则单位时间内由密度变化引起的质量流量变化为:

$$\frac{\partial \rho}{\partial t} \cdot \Delta x \cdot \Delta y \cdot \Delta z$$

根据质量守恒, 单位时间内控制体中流体质量的增量

等于通过所有控制面净流入的质量流量:

$$\frac{\partial \rho}{\partial t} \Delta x \Delta y \Delta z = - \left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] \Delta x \Delta y \Delta z$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

Continuous equation

└ There is no "-" sign with
three items, so the mass
flow rate is net outward.

单位时间内净流出控制体
的质量流量。

Jean le Rond d'Alembert

18世纪中期法国科学家达朗贝尔首先建立的

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

对于均质的不可压缩流体 (Uniform, incompressible flow)

$\rho = \text{const.}$

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

└ 散度: divergence

流体运动时, 单位体积的改变率

* 对于均质不可压缩流体的流动, 为满足质量守恒定律,
其速度矢量的散度必须为零。

3. Continuous equation in steady flow

Equivalently, exert the mass conservation
relation on control volume with arbitrary

shape and finite magnitude.

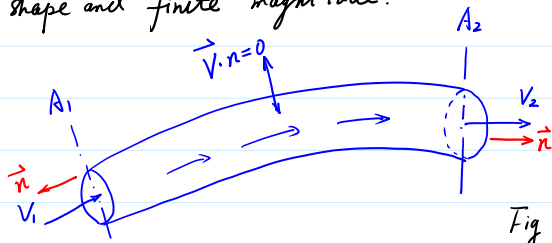


Fig 3.2

取流动中上游截面为 A_1 ，下游截面 A_2

以及两截面之间的总流壁面为控制面。

上述控制体中的流动状态若处于定常流动。

则有： $\frac{\partial \rho}{\partial t} \equiv 0$ ，

由质量守恒定律：

$$\cancel{\frac{\partial \rho}{\partial t}} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

即：由控制面净流出的质量流量为零

$$\text{For: } \nabla \cdot (\rho \mathbf{V}) = \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

$$\nabla \cdot \mathbf{V} = 0$$

因此在流动壁面上 $\mathbf{V} \cdot \mathbf{n} = u_n = 0$

而质量流量的定义式： $Q = \int_A \rho u_n dA$

则通过 A_1 流入的质量流量等于通过 A_2 流出的质量流量。

$$-\int_{A_1} \rho u_n dA_1 = \int_{A_2} \rho u_n dA_2 \quad (3.1)$$

如果采用上、下游截面的平均速度 V_1, V_2 和平均密度 ρ_1, ρ_2 来表述上述关系，则有：

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2 \quad (3.2)$$

在上游截面，平均速度 V_1 与外法向速度 $\vec{n}$ 的方向相反

所以式 (3.1) 与 (3.2) 相差一个负号。

对于不可压缩流体的管内流动，密度 ρ 为常数

式 (3.2) 可简化为： $V_1 A_1 = V_2 A_2$ (3.3)

→ Leonardo da Vinci in 1500

式 (3.3) 对于不可压缩流体的非定常运动也成立。

This states that in steady flow the mass flows entering and leaving the control volume must balance exactly.

这表明定常流动下，进入和离开控制体的质量流量必须

and leaving the control volume must balance exactly.

If the inlets and outlets are 1-dimensional, then

进入和离开控制体的质量流量必须平衡。

$$\sum_i (\rho_i A_i V_i)_{in} = \sum_i (\rho_i A_i V_i)_{out}$$

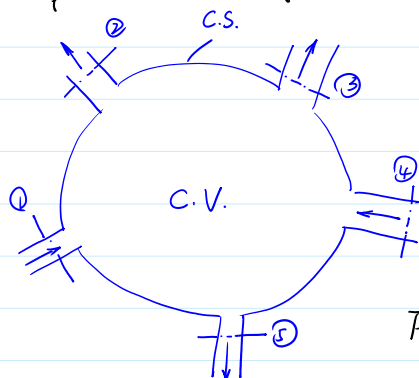


Fig 3.3

All sections i : V_{ri} approximately normal to area A_i

Section 2: uniform V_2 , A_2 , ρ_2 , etc.

if the flow in control volume is steady,

$$\text{Out-flow} = \text{In-flow}$$

$$\rho_2 V_2 A_2 + \rho_3 V_3 A_3 + \rho_5 V_5 A_5 = \rho_1 A_1 V_1 + \rho_4 A_4 V_4$$

$$\dot{m}_2 + \dot{m}_3 + \dot{m}_5 = \dot{m}_1 + \dot{m}_4$$

↳ Mass flow

$$\sum_i (\dot{m}_i)_{out} = \sum_i (\dot{m}_i)_{in}$$

If the inlets and outlets are not 1-Dimensional,

如果进出口不是一维的

one has to compute in by integration over the cross section

那么必须计算横截面的积分。

$$\dot{m}_{cs} = \int_{cs} \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

根据质量守恒, 对于 fixed control volume, incompressible

fluid, $\rho = \text{const}$,

$$\dot{Q}_{cs} = \oint_{cs} (\mathbf{V} \cdot \mathbf{n}) dA = 0, \quad V_{av} = \frac{\dot{Q}}{A} = \frac{1}{A} \int (\mathbf{V} \cdot \mathbf{n}) dA$$

↳ Volume-average velocity

Ex. 3.4

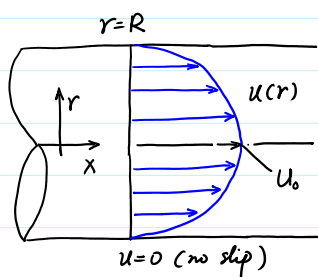


Fig 3.4

$$\rho_{av} = \frac{1}{A} \int \rho dA$$

$$(\rho \mathbf{V})_{av} = \frac{1}{A} \int \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

$$\approx \rho_{av} V_{av}$$

For steady viscous flow through a circular tube,

the axial velocity profile is given by

$$u = U_0 \left(1 - \frac{r}{R}\right)^m$$

so that u varies from 0 at the wall ($r=R$),

or no slip, up to a maximum $u=U_0$ at the

centreline $r=0$. For highly viscous (laminar)

flow $m \approx 1/2$, while for less viscous (turbulent)

flow $m \approx 1/7$, compute the average velocity if the density is constant.

Solution, $\begin{cases} V = u \\ n = i \end{cases} \Rightarrow V \cdot n = u$

Since the flow is symmetric, the differential area is taken as a circular strip,

$$dA = 2\pi r \cdot dr$$

so average velocity will be,

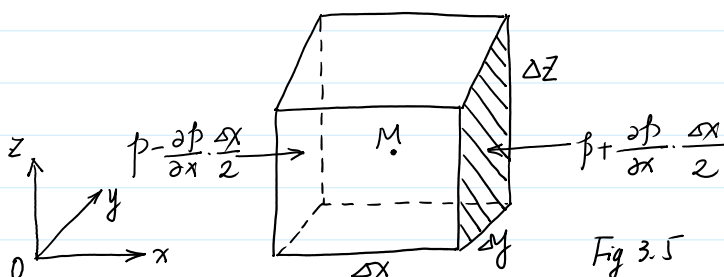
$$V_{av} = \frac{1}{A} \int u dA = \frac{1}{\pi R^2} \int_0^R U_0 \left(1 - \frac{r}{R}\right)^m 2\pi r dr$$

$$V_{av} = U_0 \frac{2}{(1+m)(2+m)} \begin{cases} m=1/2, & V_{av} \approx 0.53 U_0 \\ m=1/7, & V_{av} \approx 0.82 U_0 \end{cases}$$

§3.4 Euler Equations and integral momentum equation

对于运动的流体, 任意微小的流体系统都应该满足:

$$F = m \cdot \frac{dv}{dt}$$



① 在流场中任意取一平面六面体的微小控制体

- ② 取任意时刻 t 处于控制体内的流体为一系统
- ③ 如果忽略作用上系统上的粘性力, 那么作用在

系统上的外力有:

$$\begin{array}{ccc} \text{质量力} & + & \text{压力} \\ \underline{\underline{\quad}} & & \underline{\underline{\quad}} \\ \downarrow & & \downarrow \\ \text{作用于重心 } M & & \text{作用于系统边界面上} \end{array}$$

质量力 f_g 可以表示为 $f_g = f_x i + f_y j + f_z k$

沿 x 方向质量力的分量为:

$$\rho f_x \, dx \, dy \, dz$$

若作用在边界面上的压强为 p ,

则左边界面形心点上受到的压强为:

$$p - \frac{\partial p}{\partial x} \cdot \frac{dx}{2}$$

右边界面形心点上受到的压强为:

$$p + \frac{\partial p}{\partial x} \cdot \frac{dx}{2}$$

如果把它们当作所在面积上的平均压强: (认为作用面积
相当的注)

那么沿 x 方向作用于系统的压强合力为

$$\begin{aligned} & (p - \frac{\partial p}{\partial x} \cdot \frac{dx}{2}) \cdot dy \, dz - (p + \frac{\partial p}{\partial x} \cdot \frac{dx}{2}) \cdot dy \, dz \\ &= -\frac{\partial p}{\partial x} \, dx \, dy \, dz \end{aligned}$$

According Newton's second law of motion:

In x direction

$$\begin{aligned} F_x &= m a_x = \cancel{\rho \, dx \, dy \, dz} \frac{du}{dt} \\ &= \cancel{\rho f_x \, dx \, dy \, dz} - \cancel{\frac{\partial p}{\partial x} \, dx \, dy \, dz} \end{aligned}$$

reorganize the items,

$$du = - \frac{1}{\rho} \frac{\partial p}{\partial x} \, dx$$

reorganize the items,

$$\frac{du}{dt} = f_x - \frac{\partial p}{\partial x} \cdot \frac{1}{\rho}$$

$\xrightarrow{\quad}$ 流体质点的加速度

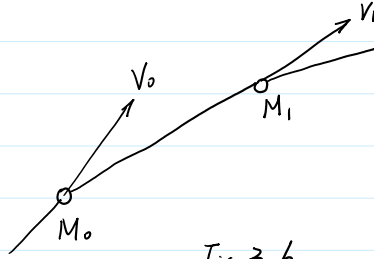


Fig. 3.6

假设任意时刻 t ，一流体质点通过空间点 $M_0(x, y, z)$ 时的速度为 $V_0(x, y, z, t)$ ，经过 Δt 时间后，该质点运动到 M_1 点， $M_1(x+\Delta x, y+\Delta y, z+\Delta z)$ ，此时速度为 $V_1(x+\Delta x, y+\Delta y, z+\Delta z, t+\Delta t)$ 。

根据加速度 a 的定义： a 等于速度 V 对时间的变化率

$$a = \lim_{\Delta t \rightarrow 0} \frac{V_1 - V_0}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta V}{\Delta t}$$

对 ΔV 进行级数展开：

$$\begin{aligned} \Delta V &= V_1(x+\Delta x, y+\Delta y, z+\Delta z, t+\Delta t) - V_0(x, y, z, t) \\ &= \frac{\partial V}{\partial t} \cdot \Delta t + \frac{\partial V}{\partial x} \cdot \Delta x + \frac{\partial V}{\partial y} \cdot \Delta y + \frac{\partial V}{\partial z} \cdot \Delta z \end{aligned}$$

$$\text{则} \quad \frac{\Delta V}{\Delta t} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} \cdot \frac{\Delta x}{\Delta t} + \frac{\partial V}{\partial y} \cdot \frac{\Delta y}{\Delta t} + \frac{\partial V}{\partial z} \cdot \frac{\Delta z}{\Delta t}$$

假设速度 V 可以表示成 $V = ui + vj + wk$ 形式

即 u, v, w 为 x, y, z 方向上的分量，则有：

$$\Delta x = u \cdot \Delta t \quad \Delta y = v \cdot \Delta t \quad \Delta z = w \cdot \Delta t$$

因此：

$$a = \frac{\Delta V}{\Delta t} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} u + \frac{\partial V}{\partial y} v + \frac{\partial V}{\partial z} w$$

$\downarrow$ $\searrow$
 固定空间点上速度对时间的变化率 由于流体质点空间位置
 称为“局部加速度” 改变而产生的速度变化
称为“对流加速度”

Recall the acceleration equation,

$$\frac{du}{dt} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x}$$

$\Downarrow$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} \quad (a)$$

对于 y, z 方向, 运用牛顿第二运动定律同样可得:

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} \quad (b)$$

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} \quad (c)$$

(a), (b), (c) 三个微分方程就是流体的运动微分方程,

又称“欧拉方程” Euler equation

上述方程只适用于理想流体 (ideal fluid, regardless of viscosity)

将三个标量方程合并:

$$\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} = \mathbf{f} - \frac{1}{\rho} \nabla p$$

$\underbrace{\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V}}_{\text{单位质量流体的惯性力}} = \underbrace{\mathbf{f}}_{\text{质量力}} - \underbrace{\frac{1}{\rho} \nabla p}_{\text{压力梯度}}$

2. Integral momentum equation

动量方程的微分形式: $\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} = \mathbf{f} - \frac{1}{\rho} \nabla p \quad (3.4)$

根据 Euler method,

$$\begin{aligned} \rho \frac{du}{dt} &= \rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} + \rho w \frac{\partial u}{\partial z} \\ &= \frac{\partial (\rho u)}{\partial t} + \frac{\partial (\rho u^2)}{\partial x} + \frac{\partial (\rho uv)}{\partial y} + \frac{\partial (\rho uw)}{\partial z} \\ &\quad - u \left[\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} + \frac{\partial (\rho w)}{\partial z} \right] \end{aligned}$$

0 (根据连续性方程)

因此根据“平2”作用于微元控制体 x 方向上的合力 F

$$= \rho u \frac{du}{dt} = \rho u \left[\frac{\partial (\rho u)}{\partial t} + \frac{\partial (\rho u^2)}{\partial x} + \frac{\partial (\rho uv)}{\partial y} + \frac{\partial (\rho uw)}{\partial z} \right]$$

因此我们可得：控制体动量变化率与施加于控制体的合力

$$F = \rho \frac{du}{dt} \Delta x \Delta y \Delta z = \left[\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} \right] \Delta x \Delta y \Delta z$$

$$= \rho f_x \Delta x \Delta y \Delta z - \frac{\partial p}{\partial x} \Delta x \Delta y \Delta z$$

因此：流体的动量变化率等于施加于流体的合力

由于： $\left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] \Delta x \Delta y \Delta z$

是通过控制面净流出的流体的质量流量。

将单位体积流体的质量 ρ 换为单位体积流体的动量 ρu

则有： $\left[\frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} \right] \Delta x \Delta y \Delta z$

→ 这是微分方程，左边的后三项

即单位时间内通过控制面进、出控制体的流体净带出的 x 方向的动量。

运动方程左边第一项： $\frac{\partial(\rho u)}{\partial t} \cdot \Delta x \Delta y \Delta z$

即：控制体内流体 x 方向的动量随时间的变化率

* 运动微分方程是针对流体系统建立的

① 系统的位置和形状随时间的改变而改变

而控制体是不变的。

② 控制体内流体动量的变化并不等于流体系统动量的全部变化。

③ 即使定常流动 $\frac{\partial(\rho u)}{\partial t} = 0$ ，当系统位置和形状时仍可能引起动量变化：这部分动量变化就是单位时间内通过控制面进、出控制体的流体净带出的动量。

④ 控制体内的动量增量 + 净带出的动量

系统动量的全部变化值

⑤ 针对控制体，运动微分方程可认为：

单位时间内控制体内流体的动量的增量与
流体通过控制面净带出的动量之和等于
作用于控制体内流体上的合外力

假设任意控制体体积为 V ，控制面为 A ，

在控制体上的动量关系为：

$$\begin{aligned} \int_V \frac{\partial(\rho V)}{\partial t} dV + \oint_A \rho V u_n dA \\ = \int_V \rho f dV - \oint_A \rho n dA \end{aligned} \quad (3.5)$$

积分形式的动量方程，简称动量方程

① 体积分：控制体内的动量变化率

② 面积分：单位时间内流体通过控制面
净带出控制体的动量，
其中 u_n 是控制面上外法线方向
的速度分量。

③ 质量力

④ 压强合力，其中 n 是控制面上外法线方
向单位矢量。

For the steady flow, consequently $\frac{\partial(\rho u)}{\partial t} = 0$

$$\text{thus, } \oint_A \rho V u_n dA = \int_V \rho f dV - \oint_A \rho n dA$$

此时系统的动量变化量就是流体通过
控制面净带出的动量。

§ 3.5 The Bernoulli Equation / frictionless flow

1. The Energy Equation

Apply first law of thermodynamics onto a fixed control volume

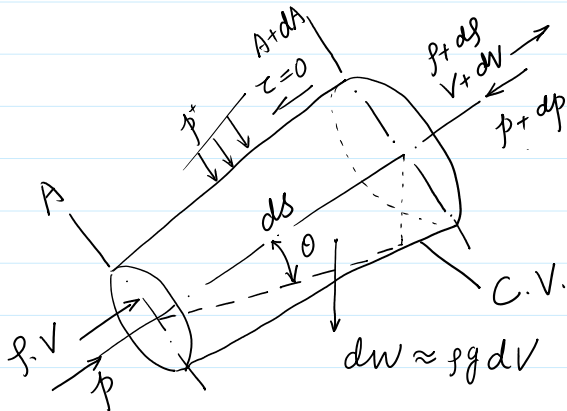
$$\begin{cases} \frac{dQ}{dt} - \frac{dW}{dt} = \frac{dE}{dt} \\ e = dE/dm \text{ (单位质量的能量)} \end{cases}$$

Recall the mass conservation equation / dm/dt

$$\frac{dE}{dt} = \frac{dm \cdot e}{dt} = \frac{d}{dt} \left(\int_{CV} e \rho dV \right) + \int_{CS} e \rho (\mathbf{V} \cdot \mathbf{n}) dA$$

Suppose the system energy could be one of the following types.

$$e = e_{\text{internal}} + e_{\text{kinetic}} + e_{\text{potential}}$$



① ρ, V, p may vary with s and time

but are assumed to be uniform over A

② Streamtube orientation θ is arbitrary.

流线与水平方向的夹角 θ
(此处与速度矢量相切的曲线)

③ elevation change $dz = ds \sin \theta$
 $\bar{L}$ length of streamline

④ Friction on the streamtube walls is shown and neglected.

1. Conservation of Mass

$$\frac{d}{dt} \left(\int_{CV} \rho dV \right) + \dot{m}_{out} - \dot{m}_{in} = 0 \quad (3.6)$$

$$\downarrow$$

$$\approx \frac{\partial \rho}{\partial t} dV + d\dot{m}$$

Here, $\dot{m} = \rho A V$, $dV \approx A \cdot ds$

$$d\dot{m} = d(\rho A V) = - \frac{\partial \rho}{\partial t} A ds \quad (3.7)$$

(上述公式并不需要无摩擦流动的假设)

2. linear momentum relation 动量守恒

$$\sum dF_s = \frac{d}{dt} \left(\int_{CV} V \rho dV \right) + (\dot{m}V)_{out} - (\dot{m}V)_{in}$$

$$\approx \frac{\partial}{\partial t} (\rho V) A \cdot ds + d(\dot{m}V)$$

where $V_s = V$ itself because s is the streamline direction.

3. If neglect shear force on the wall ($\tau=0$)

then forces are due to Pressure and Gravity

$$\begin{aligned} dF_{s, grav} &= -dW \sin \theta = -\rho g dV \sin \theta \\ &= -\gamma A ds \cdot \sin \theta \\ &= -\gamma A dz \end{aligned}$$

$$\begin{aligned} dF_{s, press} &= \frac{1}{2} dp dA - dp(A + dA) \\ &\approx A dp \end{aligned}$$

$$\text{So, } \sum dF_s = -\gamma A dz - A dp$$

$$= \frac{\partial}{\partial T} (fV) Ad_s + d(mV)$$

$$= \cancel{\frac{\partial f}{\partial t} (VA) ds} + \cancel{\frac{\partial V}{\partial t} f A ds} + \cancel{m dv} + \cancel{V dm}$$

$$[(\frac{\partial f}{\partial t} A ds + dm) V]$$

$$0 =$$

$$\cancel{\gamma A \cdot dz} + \cancel{A dp} + \frac{\partial V}{\partial t} \cancel{\gamma A ds} + \cancel{m dv} = 0 \quad / \quad p_A$$

$$g dz + \frac{dp}{\rho} + \frac{\partial V}{\partial t} ds + v dv = 0$$

Such is Bernoulli's equation for unsteady frictionless flow along a streamline

If integrate all items between any two points

$$g(z_2 - z_1) + \frac{1}{2}(V_2^2 - V_1^2) + \int_1^2 \frac{dp}{\rho} + \int_1^2 \frac{\partial v}{\partial t} ds$$

$$=$$

$$p = f(p)$$

If we consider the flow is in steady state, ($\frac{\partial V}{\partial t} = 0$)

with incompressible flow ($\rho = \text{const}$)

$$\frac{p_2 - p_1}{\rho} + g(z_2 - z_1) + \frac{1}{2}(V_2^2 - V_1^2) = 0$$

$$\text{or } \frac{p_1}{\rho} + gz_1 + \frac{1}{2}V_1^2 = \frac{p}{\rho} + gz_2 + \frac{1}{2}V_2^2 = \text{Const}$$

Above equation is for steady, frictionless, incompressible flow along a streamline. / stated in 1738 by Daniel Bernoulli

derived by Leonhard Euler.
in 1755

2. Relation between Bernoulli and Steady-flow Energy Equations

A steady-flow energy equation for a streamtube

A steady-flow energy equation for a streamtube with one inlet and one outlet,

$$\frac{p_1}{\rho} + gz_1 + \underbrace{\frac{\alpha_1 V_1^2}{2}}_{\text{kinetic-energy correction factor}} = \frac{p_2}{\rho} + gz_2 + \frac{\alpha_2 V_2^2}{2} + \underbrace{(\hat{u}_2 - \hat{u}_1 - g)}_{\substack{\text{internal energy} \\ \text{heat transfer}}} + \underbrace{W_s}_{\text{shaft work}} + \underbrace{W_v}_{\text{viscous work}}$$

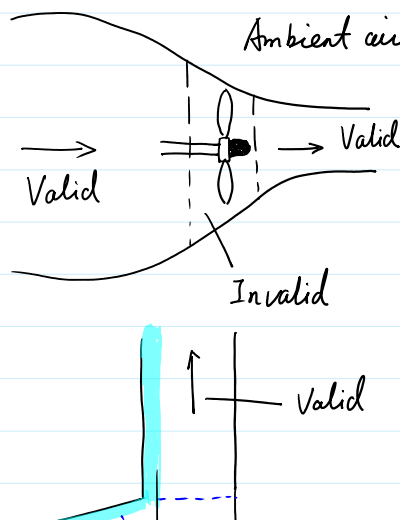
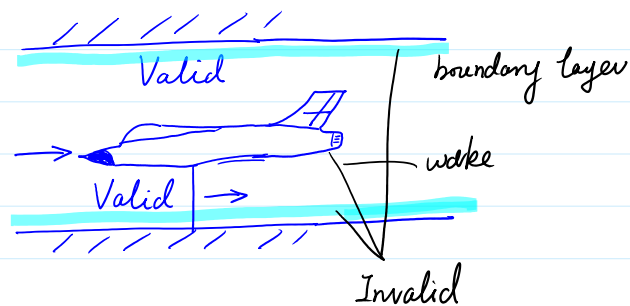
kinetic-energy correction factor: laminar 2.0, turbulent 1.04 ~ 1.11

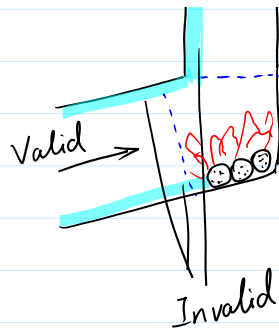
The above relation is more general than the Bernoulli equation.

Compare two equations, Bernoulli' contains even more restrictions than Steady flow Energy Eq.

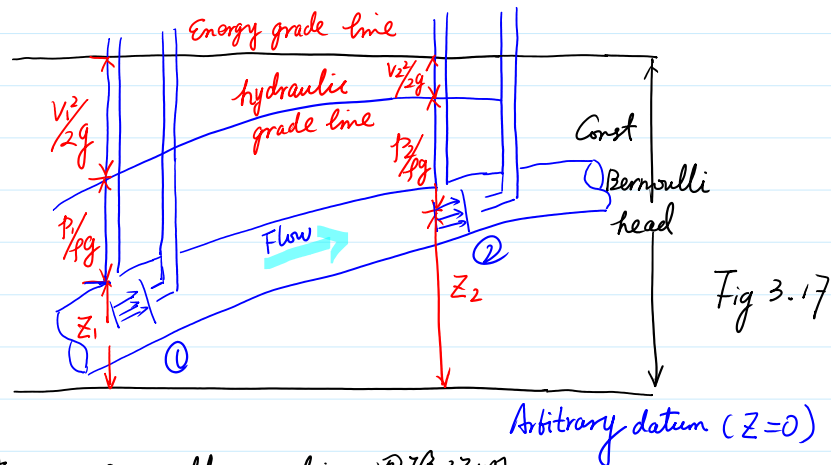
There are 6 assumptions must be satisfied.

- ① steady flow 适用范围
- ② Incompressible flow
- ③ frictionless flow
- ④ Flow along a single streamline
- ⑤ No shaft work between 1 ~ 2 / no pump or turbine
- ⑥ No heat transfer between 1 ~ 2





3. Hydraulic and Energy Grade Lines



常用的 Bernoulli equation 图解说明.

① energy grade line shows the height of Total

$$\text{Bernoulli Constant : } h_0 = Z + p/\rho + V^2/2g \quad (\text{EGL})$$

② hydraulic gradeline (HGL) shows the

height corresponding to elevation and pressure head

$$Z + p/\rho = \text{EGL} - V^2/2g \quad (\text{velocity head})$$

HGL is the height to which liquid would rise in a piezometer tube attach to the flow.

在开口的管边中, 测压管水头与水的自由表面相同

③ Fig. 3.17 说明 无摩擦流体在管内截面 1, 2 之间的 EGL 和 HGL.

测压管测量的是静压头: $Z + p/\rho$, 然后确实

HGL; 皮托滞止速度管测量总压头: $Z + p/\rho + V^2/2g$,

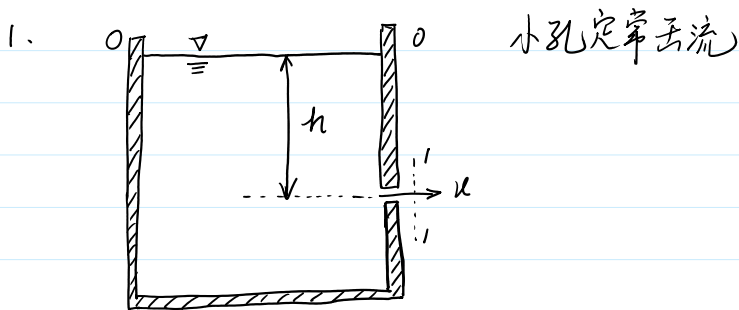
在本例中, EGL 是常数, 因此静压头随着速度降低

而升高

④ 通常, EGL 将由于摩擦损失而缓慢降低,
或者由于阀门或阻碍的存在而迅速
降低, 或者由于对外做功而降低

⑤ EGL 能通过输入功来增加,
而 HGL 一般随之速度的增加或减小
而相应降低或增加。

§ 3.6 Examples of Bernoulli equation



As known, an aperture is open at one side wall
of open-air tank, the center of aperture
is h far from free surface.

Assume that, ① The area of cross section
is much more than that of aperture

$$A_c \gg A_h \quad \text{② } dh \text{ is neglected}$$

Compute outward velocity of aperture u .

因为 h 保持不变, 所以小孔出流是定常的。

截面 (液面) 0-0 和出口截面 1-1 是渐变流
截面, 又对于大水箱, 截面 0-0 的流速可以忽略

故 $V_1 = 0$, 出口截面平均速度 $V_2 = u$ (?)

两截面上的压强均为大气压 p_a , 以孔口中心

线所在水平面为基准面, 对 0-0, 1-1 截面使用

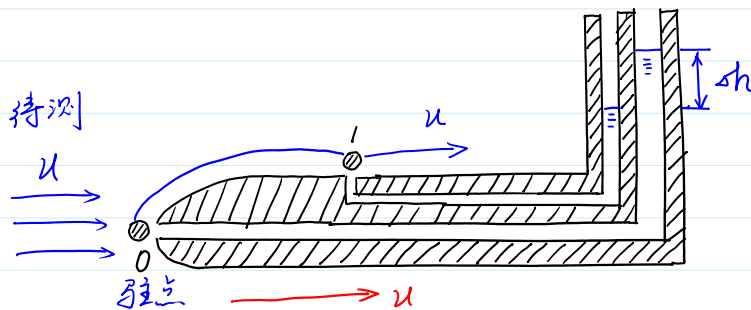
Bernoulli equation:

$$\underbrace{z_0}_{h} + \cancel{\frac{p_0}{\rho g}} + 0 = 0 + \cancel{\frac{p_1}{\rho g}} + \alpha_1 \frac{u^2}{2g} \cdot \frac{1}{2}$$

$$\text{则 } u = \sqrt{2gh}$$

2. Pitot stagnation-velocity tube

皮托管测速原理



- ① 一个椭球圆柱体, 球头中央开一孔道“0”为孔口;
- ② 在球头部后距 3~8 倍直径处沿周向开一排侧孔
将侧孔孔口称为点“1”;
- ③ 中心孔和侧孔由内套管与测压计连在一起。

假设待测流速为 u_1 ;

(a) 首先将皮托管平行于来流放置, 由于流管受到头部阻碍,
流速在头部“0”处“滞止” $u=0$, 所以此点称为“驻点”
在管口。

(b) 经过驻点的流线分支经过皮托管侧表面点 1,
速度恢复至来流速度 u_1 , 压强变为 p_1

由于两点速度不同, 故压强也不同,

两者差值使测压管中的液柱产生 Δh 的高度差

(c) 由测压管读出 Δh 后, 根据 Bernoulli equation 计算出
流速 u 。

假设皮托管直径 d 很小, 0, 1 点的高差 $\Delta h_{0,1}$ 可忽略

沿流线 0-1 列方程有:

$$\frac{p_0}{\rho g} + \cancel{gz_0} + \cancel{\frac{u_0^2}{2g}} = \frac{p_1}{\rho g} + \cancel{gz_1} + \frac{u_1^2}{2g}$$

由静力学方程: $p_0 - p_1 = \rho g \Delta h$

$$\frac{p_0 - p_1}{\rho g} = \Delta h$$

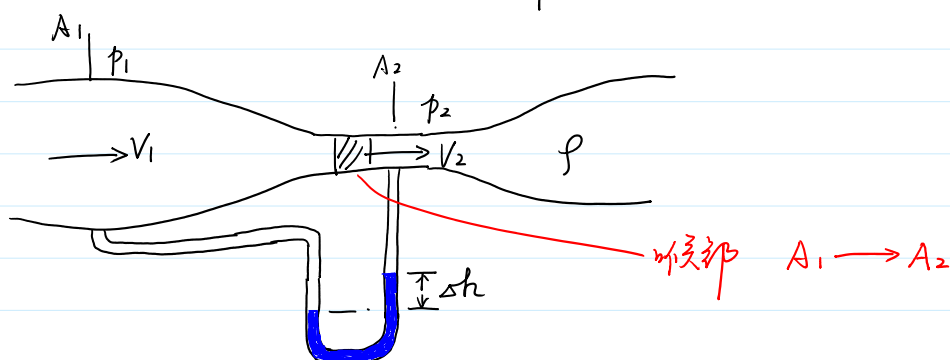
$$u_1 = \sqrt{\frac{2(p_0 - p_1)}{\rho}} = \sqrt{2g\Delta h}$$

此时: 驻点压强 p_0 , 称为总压, 或静压,

相应的测压管也分别称为总压管或静压管

$p_0 - p_1$ 通常可用 U 形管差压计来测量.

3. Venturi tube 文丘里流量计



① 文丘里流量计是一段两头粗、中间细的管道

“收缩段” “扩散段”

② 测量时把文丘里管串联到待测流量的管路中

由附带的 U 型管差压计读出液柱高差 Δh ,

从而确定管道内流动的体积流量.

假设: 截面 A_1 和截面 A_2 附近的流线均平行于管道轴线.

因此两截面之间位置差 Δh 可忽略

应用 Bernoulli equation,

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + \cancel{z_1} = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + \cancel{z_2}$$

对不可压缩流体的管道流动, 连续性方程为:
$$\begin{cases} V_1 A_1 = V_2 A_2 \\ \frac{p_1}{\rho} + \frac{V_1^2}{2} = \frac{p_2}{\rho} + \frac{V_2^2}{2} \end{cases}$$

联立整理:
$$V_2 = \sqrt{\frac{2(p_1 - p_2)}{\rho(1 - A_2^2/A_1^2)}} \quad V_1 = \frac{V_2 A_2}{A_1}$$

而截面 A_1, A_2 上的压强差可由 U 型管压差计测出,

假设测压管中工体的液体密度为 ρ' 则由静力学压强公式

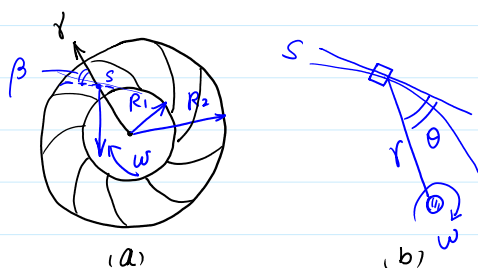
$$p_1 - p_2 = (\rho' - \rho) \Delta h$$

于是流量 Q ,

$$Q = \mu V_2 A_2 = \mu A_2 \sqrt{\frac{2g(\rho' - \rho)}{\rho(1 - A_2^2/A_1^2)}} \Delta h$$

μ — 流量修正系数, 与粘性系数和测压管精度有关
一般取 $0.95 \sim 0.98$

4. 叶轮机械中流体相对运动的佰努利方程



- ① 流体在叶轮机械中沿叶片间的通道做由内向外方向的运动,
- ② 流体随叶轮一起以等角速度沿圆周方向运动。

如图 b 的转动坐标系, 流体沿叶片通道由内向外做定常流动。

(流量一定, 角速度一定)

则取某一流线 S , 沿流线的运动方程为:

$$a_s = u \cdot \frac{\partial u}{\partial s} = f_s - \frac{1}{\rho} \frac{\partial p}{\partial s}$$

(速度不随时间变化, 只有沿流线上方向的变化)

f_s : 沿流线切向的质量力 f_s 是流体随叶轮一起旋转的离心惯性力在 s 方向上的投影。

$$f_s = \omega^2 r \cos\theta = \omega^2 \cdot r \cdot \frac{\partial r}{\partial s}$$

则有:

$$u \cdot \frac{\partial u}{\partial s} = \omega^2 r \frac{\partial r}{\partial s} - \frac{1}{\rho} \frac{\partial p}{\partial s}$$
$$\frac{\partial(\frac{u^2}{2})}{\partial s} + \frac{\partial(\frac{p}{\rho})}{\partial s} - \frac{\partial(\omega^2 \frac{r^2}{2})}{\partial s} = 0$$
$$\frac{\partial}{\partial s} \left(\frac{u^2}{2} + \frac{p}{\rho} - \frac{\omega^2 r^2}{2} \right) = 0$$

求流线 s 的积分:

$$\frac{p}{\rho g} + \frac{u^2}{2g} - \frac{\omega^2 r^2}{2g} = C$$

这就是叶轮机械中流体相对运动的伯努利方程。

以叶轮内、外缘的两点 (R_1, R_2) 应用方程:

$$\frac{p_1}{\rho g} + \frac{u_1^2}{2g} - \frac{\omega^2 R_1^2}{2g} = \frac{p_2}{\rho g} + \frac{u_2^2}{2g} - \frac{\omega^2 R_2^2}{2g}$$

假设流体总机械能: Z_1, Z_2

$$\begin{cases} Z_1 = \frac{p_1}{\rho g} + \frac{u_1^2}{2g} & \because R_2 > R_1 \\ Z_2 = \frac{p_2}{\rho g} + \frac{u_2^2}{2g} & \therefore Z_2 - Z_1 = \frac{\omega^2}{2g} (R_2^2 - R_1^2) > 0 \end{cases}$$

即: 当流体由内 $\rightarrow$ 外流动时, 叶轮外缘处的流体机械能总大于内缘处的机械能。由于旋转叶轮的叶片对流体做功, 因此流体的总机械能逐渐增加。
汽机、水轮机恰好相反。